# PENNYWISE

## Cloud-Based Personal Finance Management Application

**AWS Cloud Deployment Project**

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**Cloud Provider:** Amazon Web Services (AWS)
**AWS Region:** Asia Pacific (Sydney) – ap-southeast-2
**Deployment Date:** 9 November 2026

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# 1. Introduction

PennyWise is a web-based personal finance management application designed to help users organize and monitor their financial activities.

The application provides features for managing income, expenses, budgets, savings, transactions and financial reports.

The purpose of this project is to deploy a real-world web application using AWS cloud infrastructure while implementing cloud storage, identity management, security and application hosting.

# 2. Project Objectives

1. Build a real-world cloud-based application.
2. Utilize at least three AWS services.
3. Deploy the application using AWS.
4. Configure basic security and permissions.
5. Create an AWS architecture diagram.
6. Document the complete implementation.
7. Record a project demonstration video.

# 3. Technologies Used

- HTML – application structure.
- CSS – user interface and styling.
- JavaScript – application functionality.
- GitHub – source code management.
- Amazon EC2 – virtual server hosting.

- Amazon S3 – cloud backup storage.
- AWS IAM – identity and permissions management.
- AWS Amplify – frontend hosting and GitHub integration.
- Nginx – web server on EC2.
- Ubuntu Server – EC2 operating system.

# 4. AWS Infrastructure Implementation

## 4.1 Amazon EC2

An Amazon EC2 virtual machine was launched in the Sydney region.

Instance configuration:

- Instance name: PennyWise-Server
- Instance type: t3.micro
- Operating system: Ubuntu Server
- Public IPv4: 16.176.203.208
- Web server: Nginx

Nginx was installed and configured to serve the PennyWise application files.

## 4.2 Amazon S3

An S3 general-purpose bucket was created to store application backup files.

The application files were uploaded to the bucket, and a compressed backup archive was subsequently created.

The backup archive was uploaded to:

`backups/pennywise-backup.tar.gz`

The bucket uses private access settings with Block Public Access enabled.

## 4.3 AWS IAM

An IAM role named `PennyWise-EC2-S3-Role` was created and attached to the EC2 instance.

The role provides the EC2 instance with authorized access to the required S3 resources without embedding permanent AWS credentials in the application.

The S3 upload operation was successfully tested.

## 4.4 AWS Amplify

AWS Amplify was configured to deploy the PennyWise frontend directly from GitHub.

Configuration:

- Repository: `nahdiya-niyas/Pennywise`
- Branch: `main`
- Application name: PennyWise
- Framework: None
- Server-side rendering: Disabled

The application was successfully deployed and assigned an Amplify-hosted HTTPS URL.

**Live application:**

https://main.dz0nebtio16p.amplifyapp.com

# 5. Security Implementation

The following security measures were configured:

- EC2 security groups restrict inbound traffic to required services.
- HTTP traffic is permitted through port 80.
- SSH access is restricted to the configured client IP address, with an AWS EC2 Instance Connect rule.
- S3 Block Public Access remains enabled.
- An IAM role is used for EC2-to-S3 access.
- Application source code is maintained in GitHub.

# 6. Backup Implementation

A compressed backup archive was generated using the Linux `tar` command.

The archive was uploaded to the S3 bucket using the AWS CLI.

The successful upload was verified by listing the backup directory and confirming the presence of `pennywise-backup.tar.gz`.

This provides a cloud-based copy of the application files.

# 7. Application Deployment

The project was deployed through two hosting configurations.

**EC2 deployment:** The Ubuntu instance uses Nginx to serve the application files.

**Amplify deployment:** GitHub was connected to AWS Amplify to provide a separately hosted frontend with an HTTPS URL.

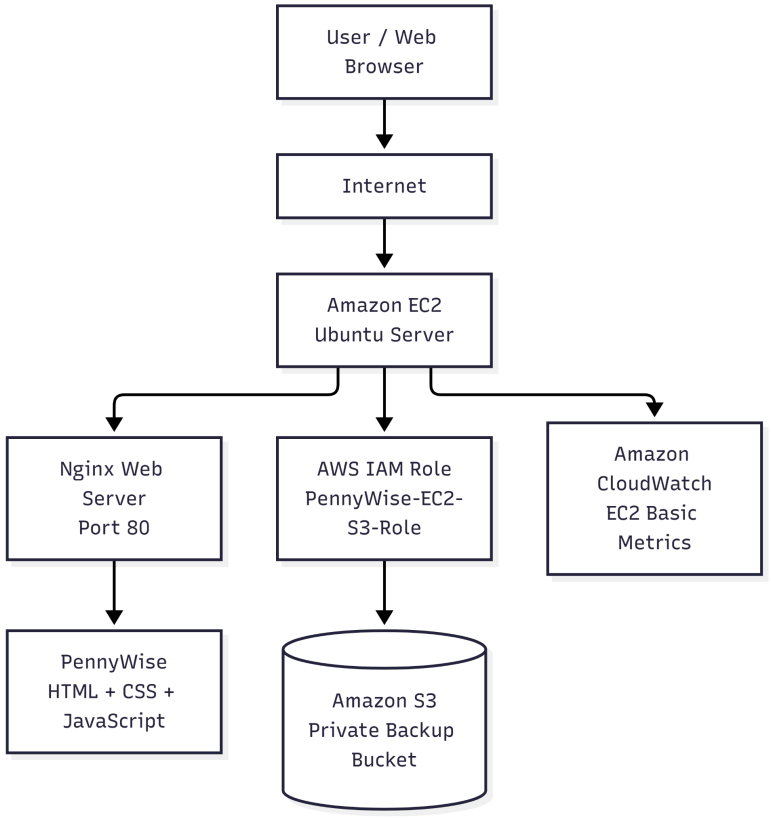
The Amplify deployment is accessible at:

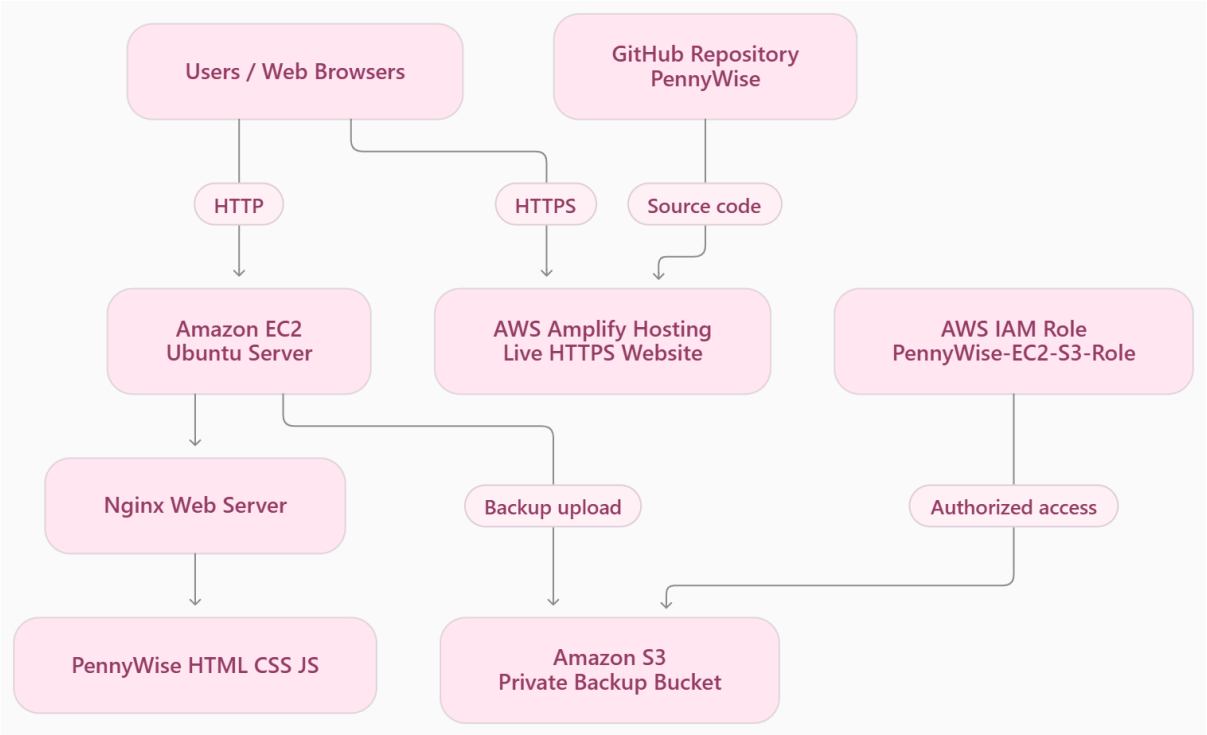
https://main.dz0nebtio16p.amplifyapp.com

# 8. AWS Architecture

The architecture consists of the GitHub source repository, Amplify hosting, the EC2 web server, the IAM role and the S3 backup bucket.

The architecture diagram should be included in this section.






# 9. Testing and Verification

The following implementation checks were performed:

| Test | Result |
|---|---|
| EC2 instance launched | Successful |
| EC2 status checks | 3/3 passed |
| Nginx service started | Successful |
| EC2 application accessible locally | Successful |
| IAM role attached | Successful |
| AWS identity verification | Successful |
| S3 application files uploaded | Successful |

| | |
|---|---|
| S3 compressed backup uploaded | Successful |
| Amplify deployment | Successful |

Additional functional testing of the Amplify application, including transaction persistence, should be completed and documented separately.

# 10. Challenges and Solutions

### EC2 SSH Connection

Initially, browser-based SSH connectivity failed. The security group configuration was reviewed and corrected, and access through the AWS terminal was subsequently established.

### GitHub Repository Access

An initial Git clone attempt used an incorrect repository path. The correct repository was subsequently accessed.

### S3 Access Permissions

The initial IAM role lacked the required S3 permissions for uploading backup objects. The permissions were corrected, and the backup upload succeeded.

### Public Website Accessibility

The EC2-hosted website was accessible from the development computer but not from another device. AWS Amplify was subsequently configured as an additional hosting solution.

# 11. Conclusion

The PennyWise project demonstrates the practical use of AWS cloud services to deploy and manage a web application.

By combining Amazon EC2, Amazon S3, AWS IAM and AWS Amplify, the project demonstrates cloud hosting, backup storage, identity-based permissions and GitHub-integrated deployment.

The implementation provides practical experience in cloud infrastructure management, deployment workflows and basic cloud security.

# 12. References

- AWS Documentation: https://docs.aws.amazon.com/
- Amazon EC2: https://aws.amazon.com/ec2/
- Amazon S3: https://aws.amazon.com/s3/
- AWS IAM: https://aws.amazon.com/iam/
- AWS Amplify: https://aws.amazon.com/amplify/
- GitHub: https://github.com/nahdiya-niyas/Pennywise
- Live Application: https://main.dz0nebtio16p.amplifyapp.com